ANKIT GOSWAMI

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Experience

Software & Test Engineering Intern **May 2025 – Jul 2025**

Robomanthan, Bangalore

- Reduced node failure rate by designing and executing structured test cases against ROS2 robotic nodes prior to deployment
- Cut manual validation effort by writing Python automation scripts for repeatable regression testing of control workflows
- Improved simulation accuracy by 25% by validating system performance against test specs and debugging root-cause issues — reflects the same debugging discipline applied to UI/frontend defects

Projects

Intrusion Detection System — *React, Tailwind CSS, ML, Python, WebSockets* GitHub

- Built a responsive network security dashboard using React (Vite) and Tailwind CSS, featuring modular KPI metric cards, real-time traffic volume charts, and active flow inspection tables
- Integrated low-latency bidirectional WebSocket communication via Flask-SocketIO, streaming live packet telemetry, bandwidth metrics, and threat alerts with sub-second update speed
- Implemented a packet sniffing and feature extraction pipeline with Scapy and Python, deploying a Random Forest ML model trained on CICIDS2017 to classify network flows as normal or malicious

Online Ruler Scale — *Astro, TypeScript, HTML5/CSS3* GitHub | Live

- Built a fully client-side, responsive on-screen ruler with three independent calibration methods — auto device-resolution detection, screen-diagonal input, and ISO/IEC 7810 credit-card-width calibration
- Delivered a mobile-first, cross-device UI covering 6+ device classes (iPhone, iPad, MacBook, Android, desktop) with a confidence-scored fallback for unrecognized displays, prioritizing UX consistency and performance
- Shipped a two-point measurement mode and shareable calibration links in TypeScript, with zero backend dependency — focused entirely on frontend performance and reusability

Skills & Competencies

Programming & Data: Python, Java, SQL, Pandas, NumPy, Pydantic, Matplotlib

AI/ML: LLMs, RAG, ML, LangChain

Developer Tools: Git, GitHub, Linux

Cybersecurity: Network Security, Network Traffic Analysis, TCP/IP, DNS, HTTP/HTTPS, Firewalls, Nmap

Frontend: HTML, CSS, JavaScript, React.js, Tailwind CSS

Certifications & Achievements

- **ROS 8 hour Hackathon Participant:** designed and tested real-time control solutions under time-boxed delivery constraints
- **Hashnova CTF Competitor:** applied systematic analytical and debugging methodology to resolve issues under time pressure

Education

BNMIT, Bangalore

Bachelor of Engineering, Electronics & Communication Engineering (E&C) *2023 – 2027*